



Microsoft Excel

Mastery Guide

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Lookups

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01

Data Validation

Control what data users can enter — prevent errors before they happen.

What is Data Validation?

Data Validation in Excel restricts the type, range, or format of data entered into a cell. It acts as a gatekeeper — keeping your spreadsheet clean and consistent.

Types of Data Validation

Type	Description	Example Setting
Whole Number	Allow only integers	Between 1 and 100
Decimal	Allow decimal numbers	Greater than 0.5
List	Dropdown from a predefined list	=A1:A10 or Male,Female
Date	Restrict to valid date range	Between 01/01/2020 and today
Time	Restrict to valid time values	Between 09:00 and 17:00
Text Length	Limit character count	Less than or equal to 50
Custom Formula	Any formula returning TRUE/FALSE	=ISNUMBER(A1)

Key Custom Formula Examples

<code>=ISNUMBER(A1)</code>	Ensures the cell contains only a number
<code>=LEN(A1)<=10</code>	Text length must not exceed 10 characters
<code>=COUNTIF(\$B\$2:\$B\$100,B2)=1</code>	Prevent duplicate entries in column B
<code>=AND(A1>=1, A1<=100)</code>	Number must be between 1 and 100
<code>=ISBLANK(A1)=FALSE</code>	Cell must not be empty
<code>=MOD(A1,5)=0</code>	Only multiples of 5 allowed
<code>=NOT(ISNUMBER(SEARCH(" ",A1)))</code>	No spaces allowed in text entry
<code>=ISODD(A1)</code>	Only odd numbers allowed

■ Pro Tip: Named Ranges for Dropdowns

Use a Named Range for dropdown lists. Name your range (e.g. 'DeptList'), then enter =DeptList in the Source box. Update the range and all dropdowns refresh automatically.

When to Use Data Validation

- Creating data-entry forms that must follow strict formats
- Building dropdown menus for consistent categorization

- Preventing invalid dates or numbers in financial models
- Ensuring unique entries (no duplicates) in ID or SKU columns
- Not needed when data is imported automatically via formulas

02

Conditional Formatting

Make your data speak visually — highlight patterns and outliers automatically.

Conditional Formatting applies visual styles (colors, icons, data bars) to cells based on rules you define — transforming raw numbers into instant visual insights without changing underlying data.

Rule Categories

Highlight Cell Rules	Greater Than, Less Than, Between, Equal To, Text Contains, Duplicates
Top / Bottom Rules	Top 10 Items, Bottom 10%, Above Average, Below Average
Data Bars	Gradient or solid fill bars proportional to cell value
Color Scales	2- or 3-color gradient spread across a range of values
Icon Sets	Arrows, traffic lights, stars and flags based on thresholds
Custom Formula	Any formula returning TRUE/FALSE — the most powerful option

Custom Formula Examples

<code>=C2>1000</code>	Highlight entire row where column C > 1000
<code>=MOD(ROW(),2)=0</code>	Alternate row shading (zebra stripes)
<code>=TODAY()-B2>30</code>	Flag items older than 30 days
<code>=AND(C2>0,D2="Pending")</code>	Highlight rows: sales > 0 AND status is Pending
<code>=WEEKDAY(A2,2)>5</code>	Highlight weekend dates (Saturday/Sunday)
<code>=ISBLANK(A2)</code>	Highlight empty cells in column A
<code>=RANK(B2,B\$2:B\$100)<=3</code>	Highlight top 3 values in column B
<code>=C2/SUM(C\$2:C\$100)>0.1</code>	Cells contributing more than 10% of total
<code>=LEN(TRIM(A2))<>LEN(A2)</code>	Detect leading or trailing spaces in text
<code>=A2=A1</code>	Highlight duplicate consecutive cell values

■ Pro Tip: Relative vs Absolute References in CF Rules

When formatting entire rows, anchor the column with \$ but leave the row relative: `=C2>1000`. This lets each row evaluate independently while always checking column C.

When to Use Conditional Formatting

- Dashboard KPIs — red/amber/green traffic-light indicators

- Sales data — highlight best and worst performers instantly
- Project tracking — overdue tasks or upcoming deadlines
- Financial models — flag negative values or budget overruns
- Data cleaning — spot blank cells, duplicates, or outliers
- Avoid on very large datasets (100k+ rows) — can slow Excel significantly

03

Lookup Functions Overview

Find, retrieve, and cross-reference data across your spreadsheets.

Excel's lookup functions let you search for a value in one dataset and return related information from another — the backbone of any data-driven spreadsheet.

Lookup Function Comparison

Function	Direction	Complexity	Best Use Case	Key Limitation
VLOOKUP	Vertical ↓	Simple	Find by row key	Left-to-right only
HLOOKUP	Horizontal →	Simple	Find by column header	Top-to-bottom only
LOOKUP	Both	Simple	Sorted data search	Requires sorted data
INDEX	Both	Moderate	Return cell at position	Needs row+col number
MATCH	Both	Moderate	Find position of value	Returns position, not value
INDEX+MATCH	Both	Moderate	Most flexible lookup	Two functions combined
XLOOKUP	Both	Simple	Modern replacement	Excel 365/2019+ only

MATCH Function

Returns the relative position of a value in a range.

<code>=MATCH(lookup_value, lookup_array, [match_type])</code>	Core syntax
<code>=MATCH("Alice", A2:A100, 0)</code>	Exact match — returns row position of 'Alice'
<code>=MATCH(500, B2:B50, 1)</code>	Largest value <= 500 (ascending sort required)
<code>=MATCH(500, B2:B50, -1)</code>	Smallest value >= 500 (descending sort required)

INDEX Function

Returns the value at a specific row/column intersection.

<code>=INDEX(array, row_num, [col_num])</code>	Core syntax
<code>=INDEX(A1:D100, 5, 3)</code>	Value from row 5, column 3 of A1:D100
<code>=INDEX(B2:B100, 10)</code>	10th value from a single-column range

INDEX + MATCH — The Power Combination

<code>=INDEX(C2:C100, MATCH("Alice", A2:A100, 0))</code>	Find Alice in col A, return her salary from col C
<code>=INDEX(A2:D100, MATCH("Q3", A2:A100, 0), MATCH("Revenue", A1:D1, 0))</code>	2D lookup: find row by Q3, column by Revenue header

```
=INDEX($B$2:$B$100, MATCH(MAX($C$2:$C$100), $C$2:$C$100, 0))
```

Name of the person with the highest score

■ Pro Tip: Why INDEX+MATCH Beats VLOOKUP

INDEX+MATCH can look left, is unaffected by column insertions, and handles large datasets faster than VLOOKUP. Consider it your go-to for complex lookups.

04

VLOOKUP — Vertical Lookup

Search down a column and return a value from the same row.

Syntax

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

Parameter	Required?	Description
lookup_value	Required	The value to find (text, number, or cell ref)
table_array	Required	Range containing data — first column must hold lookup values
col_index_num	Required	Column number in table_array to return (1 = first column)
range_lookup	Optional	FALSE = exact match (recommended) TRUE = approximate match

VLOOKUP Examples — Basic to Advanced

=VLOOKUP(A2, \$D\$2:\$G\$100, 2, FALSE)	Exact match: look up A2 in D2:G100, return column 2
=VLOOKUP("EMP001", EmployeeTable, 3, FALSE)	Lookup employee ID in named range, return 3rd col (department)
=VLOOKUP(B5, PriceList, 2, FALSE)	Look up product in B5 in price list, return price from col 2
=VLOOKUP(A2, \$B\$2:\$E\$50, 4, TRUE)	Approximate match — for tax brackets or grade ranges (data must be sorted)
=IFERROR(VLOOKUP(A2, \$D\$2:\$G\$100, 2, FALSE), "Not Found")	Show 'Not Found' instead of #N/A error
=IFERROR(VLOOKUP(A2, Sheet2!\$A:\$D, 3, FALSE), "-")	Cross-sheet VLOOKUP — look up data from Sheet2
=VLOOKUP(A2&B2;, \$D\$2:\$F\$100, 3, FALSE)	Multi-criteria: concatenate two keys (requires a helper column)

Grade Calculation — Approximate Match Example

Min Score	Grade		Student Score	Formula Result
0	F		45	=VLOOKUP(D2, \$A\$2:\$B\$6, 2, TRUE) → F
50	D		72	=VLOOKUP(D3, \$A\$2:\$B\$6, 2, TRUE) → C
60	C		85	=VLOOKUP(D4, \$A\$2:\$B\$6, 2, TRUE) → B
70	B		95	=VLOOKUP(D5, \$A\$2:\$B\$6, 2, TRUE) → A
80	A			

Common VLOOKUP Errors & Fixes

Error	Cause	Fix
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#N/A	Value not found in first column	Use IFERROR(); check for spaces with TRIM()
#REF!	col_index_num exceeds table width	Count columns; col_index must be ≤ table width
#VALUE!	col_index_num < 1	col_index must be a positive integer
Wrong result	range_lookup=TRUE on unsorted data	Always use FALSE for exact match

■ Pro Tip: Always Use Absolute References (\$)

Always lock your table_array with absolute references (\$D\$2:\$G\$100). When copying the formula down, relative refs shift — causing wrong results or #REF! errors. Press F4 after selecting the range to toggle absolute reference.

When to Use VLOOKUP

- Employee ID → Name / Department / Salary lookup
- Product SKU → Price / Description lookup
- Tax bracket or grade range calculations (approximate match)
- Merging two tables using a common key column
- Avoid when the lookup column is NOT the first column of your range
- Avoid on very large datasets — use INDEX+MATCH or Power Query instead

05

HLOOKUP — Horizontal Lookup

Search across a row and return a value from the same column.

HLOOKUP works like VLOOKUP but searches horizontally across the first row, then returns a value from a specified row below. Ideal when your headers run across columns rather than down rows.

Syntax

`=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])`

Parameter	Required?	Description
lookup_value	Required	Value to search for in the FIRST ROW of table_array
table_array	Required	Range — first row must contain the lookup header values
row_index_num	Required	Which row number to return (1 = header row itself)
range_lookup	Optional	FALSE = exact match TRUE = approximate (sorted first row)

HLOOKUP Examples

<code>=HLOOKUP("Q3", \$B\$1:\$E\$10, 3, FALSE)</code>	Find 'Q3' in row 1, return row 3 of same column — exact match
<code>=HLOOKUP("Revenue", \$A\$1:\$Z\$5, 2, FALSE)</code>	Look up 'Revenue' header, return row 2 value (e.g. Jan revenue)
<code>=HLOOKUP(D1, MonthlyData, 4, FALSE)</code>	Dynamic: D1 contains a month name, returns row 4 data
<code>=HLOOKUP("Bonus", \$A\$1:\$L\$8, MATCH(A2,\$A\$1:\$A\$8,0), FALSE)</code>	Combine MATCH for a dynamic row_index_num
<code>=IFERROR(HLOOKUP(A1,\$B\$1:\$Z\$10,2,FALSE),0)</code>	Return 0 instead of #N/A if column header is not found
<code>=HLOOKUP("Target", Summary!\$A\$1:\$M\$6, 5, FALSE)</code>	Cross-sheet HLOOKUP referencing the Summary tab

Example Dataset — Monthly Sales by Region

Region	Jan	Feb	Mar	Q1 Total
North	45,200	51,300	48,700	145,200
South	38,900	42,100	55,600	136,600
East	62,400	58,200	70,100	190,700
West	29,800	33,500	37,200	100,500

<code>=HLOOKUP("Feb", \$A\$1:\$E\$5, 3, FALSE)</code>	South's Feb sales = 42,100 (row 3 = South)
<code>=HLOOKUP("Q1 Total", \$A\$1:\$E\$5, 4, FALSE)</code>	East's Q1 Total = 190,700 (row 4 = East)

■ Pro Tip: When HLOOKUP Makes Sense

HLOOKUP is perfect for financial models where time periods (Jan, Feb, Mar...) are column headers and metrics are rows. If you use HLOOKUP frequently, consider whether transposing your data to a vertical layout would make analysis easier.

When to Use HLOOKUP

- Financial reports with months/quarters as column headers
- Comparison matrices where categories are columns
- Score/rating tables with criteria along the top row
- Configuration tables where settings are column names
- Avoid when data is structured vertically — use VLOOKUP instead
- Not ideal for large datasets — INDEX+MATCH gives better performance

06

Comparison & Decision Guide

Choose the right lookup tool for every situation.

VLOOKUP vs HLOOKUP vs INDEX+MATCH vs XLOOKUP

Feature	VLOOKUP	HLOOKUP	INDEX+MATCH	XLOOKUP
Search Direction	Vertical ↓	Horizontal →	Both ⇕↔	Both ⇕↔
Lookup Column Position	Must be 1st	Must be 1st row	Any position	Any position
Return Direction	Right only →	Down only ↓	Any direction	Any direction
Multiple Results	First only	First only	Array capable	Built-in
If Not Found	Needs IFERROR	Needs IFERROR	Needs IFERROR	Built-in param
Performance	Moderate	Moderate	Fast	Fast
Excel Version	All versions	All versions	All versions	365/2019+ only
Ease of Use	Easy	Easy	Moderate	Easy-Moderate

Decision Guide — Which Function Should I Use?

Data is vertical AND lookup column is first?	→ Use VLOOKUP
Data is horizontal AND lookup row is first?	→ Use HLOOKUP
Lookup column is NOT first / need left lookup?	→ Use INDEX+MATCH
Need not-found handling built in?	→ Use XLOOKUP (if available)
Complex 2-way lookup (row AND column)?	→ Use INDEX+MATCH (2D)
Simple range lookup on sorted data?	→ Use VLOOKUP with TRUE

Bonus: XLOOKUP — The Modern Standard (Excel 365+)

<code>=XLOOKUP(lookup_value, lookup_array, return_array, [not_found], [match_mode], [search_mode])</code>	Full syntax — most parameters are optional
<code>=XLOOKUP("Alice", A2:A100, C2:C100)</code>	Find Alice in col A, return salary from col C
<code>=XLOOKUP("Alice", A2:A100, B2:D100)</code>	Return multiple columns at once — not possible with VLOOKUP
<code>=XLOOKUP(A2, A2:A100, B2:B100, "Not Found")</code>	Built-in not-found message — no IFERROR wrapper needed

`=XLOOKUP(A2, C2:C100, B2:B100)`

Left lookup — return_array can be to the LEFT of lookup_array

Best Practices Summary

✓ Always use exact match (FALSE/0)	Approximate match silently returns wrong results if data is not perfectly sorted ascending.
✓ Lock table references with \$	Use \$A\$2:\$D\$100 — formulas shift when copied, causing incorrect lookups.
✓ Wrap lookups in IFERROR	=IFERROR(VLOOKUP(...), 'Not Found') prevents ugly #N/A errors in reports.
✓ Use Named Ranges	Name your lookup tables (e.g. PriceTable) for readable, maintainable formulas.
✓ Prefer INDEX+MATCH for large data	It searches only one column/row vs VLOOKUP which scans the entire table array.
✓ Migrate to XLOOKUP when possible	Simpler syntax, built-in error handling, supports left lookups and multiple results.

■ Future-Proofing: Start Migrating to XLOOKUP

Microsoft officially recommends XLOOKUP as the replacement for both VLOOKUP and HLOOKUP in Excel 365. If your organization uses Excel 365 or Excel 2021+, start adopting XLOOKUP — it's simpler, more powerful, and handles edge cases without extra wrapper functions.